

2.10.46

EE25BTECH11017 - Mahesh Chollangi.

Question:

Let $\mathbf{V} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{W} = \mathbf{i} + 3\mathbf{k}$. If $\mathbf{U}$ is a unit vector, then the maximum value of the scalar triple product $[\mathbf{U} \ \mathbf{V} \ \mathbf{W}]$ is ...

Solution:

Let

$$\mathbf{V} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} U_x \\ U_y \\ U_z \end{pmatrix}, \quad \|\mathbf{U}\| = 1 \quad (1)$$

The scalar triple product is:

$$S = \det \begin{pmatrix} U_x & 2 & 1 \\ U_y & 1 & 0 \\ U_z & -1 & 3 \end{pmatrix} \quad (2)$$

Expanding the determinant, we get:

$$S = U_x \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} - U_y \det \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} + U_z \det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \quad (3)$$

$$= U_x(1 \times 3 - 0 \times (-1)) - U_y(2 \times 3 - 1 \times (-1)) + U_z(2 \times 0 - 1 \times 1) \quad (4)$$

$$= 3U_x - 7U_y - U_z \quad (5)$$

As $\mathbf{U}$ is unit vector,

$$U_x^2 + U_y^2 + U_z^2 = 1 \quad (6)$$

For a vector $\mathbf{A} = \begin{pmatrix} 3 \\ -7 \\ -1 \end{pmatrix}$, the maximum value of scalar product $\mathbf{U}^\top \mathbf{A}$ subject to $\|\mathbf{U}\| = 1$ is $\|\mathbf{A}\|$.

Calculate

$$\|\mathbf{A}\| = \sqrt{3^2 + (-7)^2 + (-1)^2} \quad (7)$$

$$= \sqrt{9 + 49 + 1} = \sqrt{59} \quad (8)$$

Hence,

$$\max [\mathbf{U} \ \mathbf{V} \ \mathbf{W}] = \boxed{\sqrt{59}} \quad (9)$$

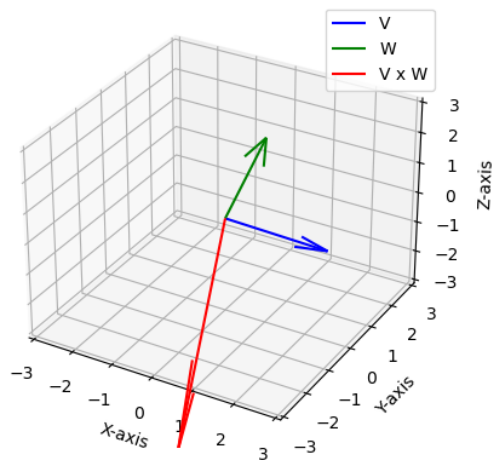
Vectors V , W and Cross Product $V \times W$ 

Fig. 0.1: 3D Plot